

SymPy Bug Card

Bug 60: LerchPhi Left Limit Returns Divergent Endpoint

Task. Compute the left-hand limit of the Lerch transcendent:

$$\lim_{x \rightarrow 1^-} \text{lerchphi}(x, 1, 1).$$

SymPy's answer. *SymPy* returns the unevaluated endpoint

$$\text{limit}(\text{lerchphi}(x, 1, 1), x, 1, \text{dir} = -) = \text{lerchphi}(1, 1, 1).$$

Correct answer. The mathematically correct result is

$$+\infty.$$

Explanation. For $0 < x < 1$,

$$\text{lerchphi}(x, 1, 1) = \sum_{n=0}^{\infty} \frac{x^n}{n+1} = \frac{-\log(1-x)}{x},$$

so the left-hand limit diverges to $+\infty$. The returned endpoint represents the harmonic series at $x = 1$, not a finite limit value.

Likely root cause. The diagnosis traces the limit through Gruntz, where `mr_v_leadterm` treats $\text{lerchphi}(1 - 1/z, 1, 1)$ as having coefficient $\text{lerchphi}(1, 1, 1)$ and exponent 0, instead of exposing the logarithmic divergence available through expansion. This is a new instance of a known failure mode.

Artifact (GitHub).

[bug-report/bug-060-lerchphi-left-limit-returns-divergent-endpoint/](#)